

Analysis of customer reviews

Group :

Name :

Todo

- **Translation, working with Google sheets, everyone takes a file**
- **Do not make K means after ACP**
- **Presentation of scientific articles or concepts**
 - Chatbot (chatGPT)
 - Text generation
 - Embedding sentences (USE)
 - Text summary
 - Translation of texts
 - Image generation from text (Dall-E)
 - Transformers
 - BERT and variants
- **Spell checker**
- **Choice of themes**

Scoring/mark

- **Data exploration and cleansing: 2 points (negative points if not carried out properly)**
 - Statistics about the dataset (nb of reviews, per insurer, per insurance product, average nb of stars...)
 - Frequent words (and n-grams)
 - Correct spelling: 2 points
 - Summary, translation, question answering, and generation: 2 points
- **Unsupervised learning (2 points per model)**
 - **Topic modelling** and lists of topics (group work): 2 points (negative points possible)
 - Anomaly detection
- **Embedding to identify similar words and enrich the list of themes: 2 points (possible negative points)**
 - Word2vec training: 2 points, GloVe: 2 points
 - Embedding visualisation with matplotlib and Tensorboard: 2 points
 - Implementation of the Euclidean distance or cosine: 1 point
- **Supervised learning, each model well done and well presented: 2 points (possible negative points)**
 - Sentiment detection (multiclass, or binary classification): 2 points (negative points possible)
 - TF-IDF and classic ML
 - Basic model with an embedding layer (viewing embeddings with Tensorboard: 1 extra point)
 - Model with pre-trained embeddings (viewing embeddings with Tensorboard: 1 extra point)
 - **USE** (Universal Sentence Embedding) or equivalent, RNN **LSTM**, **CNN**, **BERT** or other models on Hugging Face
- **Interpretation of results (possible negative points)**
 - Error analysis: 1 point (or each model), negative points if not done (get special cases)
 - shap : 2 points
- **Clarity of presentation: 2 points (negative points possible)**
- **Streamlit app (link to specify in the report)**
 - Search (with BM 25, embedding)
 - Presentation of the analysis of one review
- **Post on LinkedIn and get 200 likes/comments (presentation of interesting results)**

Example of a table of contents

- **Part 1**
 - Part 1.1.
 - Part 1.2.
- **Part 2**

Example of a table of contents

1 | General presentation of the project and data

1.1 | Presentation of the problem

1.2 | Data presentation

1.3 | Data exploration and cleansing

3 | Feature engineering and modelling

2.1 | Creating new variables

2.2 | Building and comparing predictive models

4 | Interpretation of results

3.1 Importance of variables

3.3 | Conclusion on the use of models

Purpose and context

- **Context :**
 - Customer satisfaction surveys have been carried out, and an analysis of open comments has been carried out.
- **Objectives:**
 - Identifying themes in comments
 - Understanding the impact of themes on customer satisfaction levels
- **Techniques used :**
 - Carrying out an exploratory analysis
 - identify themes

1 | Presentation of the project's objective and data

- **General file structure**
- **Presentation of variables**
- **Analysis/presentation of target variables**
- **Presentation of explanatory variables/features/input**

1.1 | Objective and context

Objectives

- Understanding products and insurers and their average ratings
- Viewing changes in scores over time

Presentation of the objective

- **Non-technical presentation: objective, usefulness,**
- **Technical presentation: supervised learning, unsupervised learning, machine learning, deep learning**

- **Project 1 : information retrieval, give relevant hotels (via the reviews)**
- **Project 2 :**

- **Examples of NLP issues using the Open Food Facts database**
 - Segmenting products with ingredients
 - Apply word2vec to ingredients: display similar ingredients
 - Apply global average pooling for products, and search for similar products

Background Google & Amazon Opinions (1/2)

In Google and Amazon reviews, frequent words are quoted to enable users to select the reviews concerned.

- Stopwords to be reprocessed
- Several words together (bigrams, trigrams) to be detected

→ do better :

- for each keyword, we give an associated rating (either the rating for the entire review, or better still, an assessment of the level of satisfaction for this particular keyword. This is because a review can cover several themes or keywords, with different levels of satisfaction, and the final score is overall).
- Search for similar terms: play area and playroom

4,0 ★★★★★ 569 avis
Les avis n'ont pas été validés. ⓘ

Thèmes qui reviennent souvent

Tous les avis espace de jeux 53 parents 52 sous sol 51 concept 43 salle de jeux 41

écrans 23 service 23 menu enfant 18 caméras 14 surveiller 14

Trier par

Avis les plus pertinents Les plus récents Les meilleurs Les moins bons

Avis Google

Tous les avis plats 45 service 16 prix 16 voyage 6

formule 5 fondue 4 bo bun 3 parfumés 3 goûté 3

Lire des commentaires qui mentionnent

lave vaisselle très simple grandes assiettes fruits et légumes

mini lave-vaisselle laver les fruits verres et des couverts litres eau

très bien très pratique lave-vaisselle très lavage bruit

Background Google & Amazon Opinions (2/3)

Focus on stop words

The "stop words" seem to have been cleaned up.

For "very", its usefulness may be questionable.

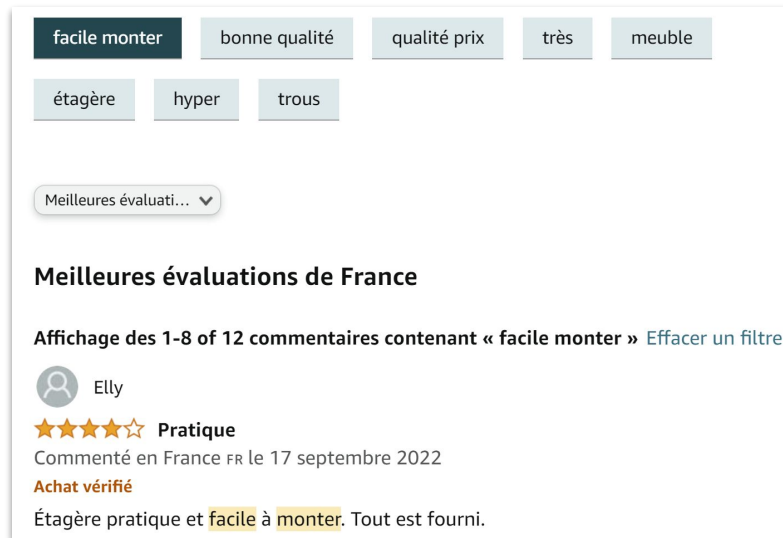
This product is a piece of furniture/bookcase, so the word "furniture" could be considered a stop word.

Lessons learned: even though Google and Amazon are at the cutting edge of NLP R&D, we can see that for certain use cases, simple techniques are still used.

In practice, there are two types of stop words

- stop words génériques
- stop words by domain

And frequency helps to highlight most of them...



The screenshot shows a product review interface. At the top, there are filter buttons: 'facile monter' (selected), 'bonne qualité', 'qualité prix', 'très', and 'meuble'. Below these are more filters: 'étagère', 'hyper', and 'trous'. A dropdown menu shows 'Meilleures évaluati...'. The main heading is 'Meilleures évaluations de France'. Below that, it says 'Affichage des 1-8 of 12 commentaires contenant « facile monter »' with a link 'Effacer un filtre'. The first review is from a user named 'Elly' with a 4-star rating and the title 'Pratique'. It says 'Commenté en France FR le 17 septembre 2022' and 'Achat vérifié'. The review text is 'Étagère pratique et facile à monter. Tout est fourni.' with 'facile à monter' highlighted in yellow.

Context Google & Amazon Opinions (3/3)

On Amazon, themes seem to be detected automatically from reviews. They are then proposed for rating when a customer leaves a review. Themes can be deleted if the customer does not consider them relevant enough.

At the end, we have notes for specific themes.

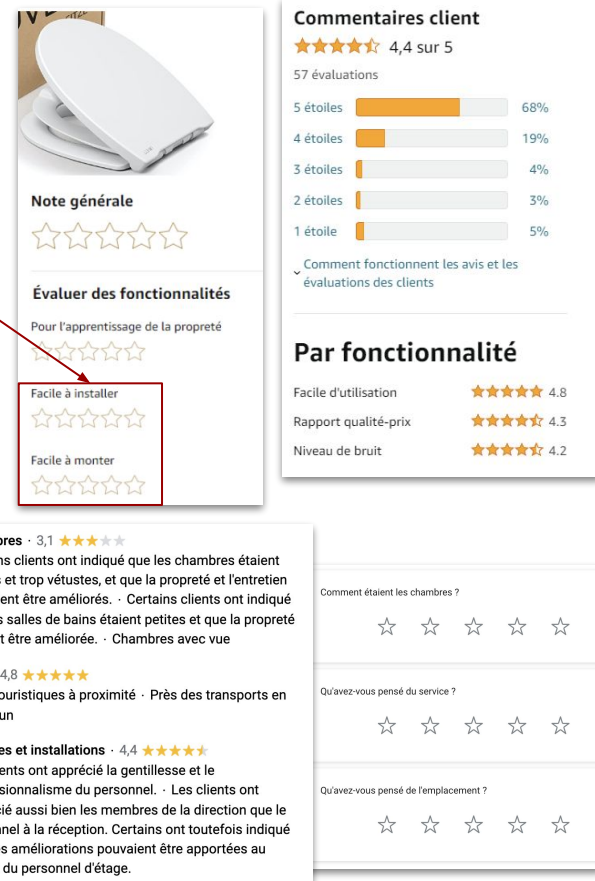
Google has the same feature, but apparently only for hotels. And sometimes whole sentences are suggested.

The following conclusions can be drawn:

- Creating themes is not easy and cannot be automated.
- The themes must be defined by experts. They are not the same for all products.

The proposed criteria are undoubtedly probabilistic. And the themes are independent. Because "easy to install" and "easy to assemble" are similar.

These themes, in the form of sentences, are surely written beforehand and then detected using classification algorithms.



1.2 | Data presentation

Objectives

- Understanding products and insurers and their average ratings
- Viewing changes in scores over time

Data presentation

- **Variables**
 - Context of publication of notices (author, date of publication)
 - Context of the insurance product (type of product, insurer)
 - Customer reviews
 - Translation of the notice into English
 - Related note
- **Present the data in categorical macro format (in a pyramid) to make it clear.**
- **Describe some anomalies and their causes**

Data presentation (project 1)

- **Variables**
 - text
 - hotel (id and name)
 - rating (“service”, “cleanliness”, “overall”, “value”, “location”, “sleep quality”, “rooms”)
- **Present the data in categorical macro format (in a pyramid) to make it clear.**
 - number of hotels
 - average rating (distribution, density or histogram)
 - average text length (distribution, density or histogram)

Translation

- **For Project 1**
 - Language detection
 - Then for French, you can translate into English
- **Hugging face: <https://huggingface.co/tasks/translation>**
- **Translation with Google Sheets, googletranslate function**
 - Everyone takes a file, demo to do
 - Correcting texts to obtain correct translations
 - https://drive.google.com/drive/folders/1ktwly8jQJraANsPAM-pjsyGIVFu_T_xM?usp=sharing

Examples of observations

- Give some examples of observations
- Highlighting anomalies, spelling mistakes and translation errors

Metrics used

- **Metrics (average, density) before cleaning**
- **Metric after cleaning**
- **Count the number of lines deleted and the reasons why**

2 | Exploration/Transformation/ feature engineering

- Data cleansing
- Simple exploration and cleaning
- Word embedding (self-supervised)
- Other NLP tasks

Information retrieval system

- **Input : query**
- **Output**
 - multiple answers

Example for review

- **Input : one review, with a rating (example 4)**
- **Output**
 - Exemple : 5 reviews with their ratings (4, 4, 5, 5, 4), then we calculate the MSE, here low, so good system of IR
 - Example : 5 reviews with ratings (1, 1, 1, 1, 1), then bad IR system

Approaches of information retrieval for P1

Information to present

- Two dataframes : review_df and hotel_df (average rating by hotel)
 - hotel_df : average rating, reviews number, summary of the reviews
- Filter the dataframe (with different techniques)
- (Group by hotels)
- Order the rows according to a score (with option to group by hotels)
- Order by rating (review or hotel)
- Order by relevance score (from information retrieval technique)
- Show different results of different information retrieval techniques below

Different approach for information retrieval

- Regex keyword detection (or LV for each token)
- TF IDF / BM 25
 - Compute score
 - Sort
- Semantic search : embedding for each review (different embeddings techniques)
 - Create embedding
 - similarity calculation and sort

Approaches of information retrieval for P1

Evaluation of information retrieval

- for some reviews, you consider them as query
- then you get similar reviews
- then you get the top n, and then you can calculate the MSE
- at the end, you can compare the MSE for different techniques (for each rating)
- BN : this evaluation technique is not adapted, since one review may only talk about one criterion (relevant to one rating, for example only “rooms”), then it is not correct to use other rating to do the evaluation)

Hotel similarity search

- for some reviews, you consider them as query
- then you get similar reviews
- then you can get the most similar hotel

2.1 | Data cleansing

Objectives

- Understanding products and insurers and their average ratings
- Viewing changes in scores over time

Feature engineering for P1

- TF IDF / BM 25
-

Simple cleaning

- **Experience and publication dates**
 - <https://pbpython.com/text-cleaning.html>
- **Simple word/token count**
- **Basic spelling correction, initially for frequent words**
- **Handling words with accents**
- **Analysis of n-grams**

Transformation/cleaning of variables according to their nature

- **Numerical variables**
- **Categorical variables**
 - If the number of categories is less than 6, count the number...
 - If there are more than 6 categories, group them together...
- **Text variables**
 - BoW
 - TF-IDF
 - Word2vec

Summary of cleaning work carried out

- **Restatement of dates**
- **Removal of notices with missing text**
- **Text cleaning**
 - Removing stop words (empty words)
 - Deleting special characters
 - Replacement of certain words ("customer service" by "customer_service", study to find bigrams or trigrams).
- **Export the cleaned file so that you don't have to do it again for future sessions. This file must contain the following columns**
 - Date of publication of the notice
 - Month of experience (first day of the month)
 - Notice in French without cleaning
 - Notice in French with cleaning 1 for certain models
 - Avis en français avec nettoyage 2 pour d'autres modèles
 - The same applies to notices translated into English

2.2 | Data mining

Objectives

- Understanding products and insurers and their average ratings
- Viewing changes in scores over time

Global data mining

- **Number of reviews and average rating by product, insurer, date, etc.**
- **Number of words in a notice, depending on the rating**
- **Create models by product if necessary following exploration**

Exploring opinions

- **Count and average score (and how they change over time) as a function of the following variables**
 - Product
 - Insurer
- **As far as comments are concerned, we need to analyse them on a product-by-product basis, because you might think that the themes would not be the same. In practice, different departments within the company deal with them, and we need to have conclusions for each product.**
- **Constitution of n-grams (several words together)**
 - As we can see on Google and Amazon, and we can notice a subtlety according to examples seen: on Google, the n-grams are constituted before the deletion of the empty words ("espace de jeux" with "de" in it), and on Amazon, the n-grams are identified after the deletion of the empty words ("facile à monter" becomes "facile monter").

2.4 | Other NLP tasks

Objectives

- Fine-tuning the themes

Task summary

- **Spelling correction**
- **Text summary**
- **Text generation**
 - By product and by insurer, compare and highlight differences
- **Sentiment detection with transfer learning**
 -
- **Translation: testing several packages**
 - In practice, Google Translate can be used (see data presentation)

3 | Unsupervised learning /self-supervised learning

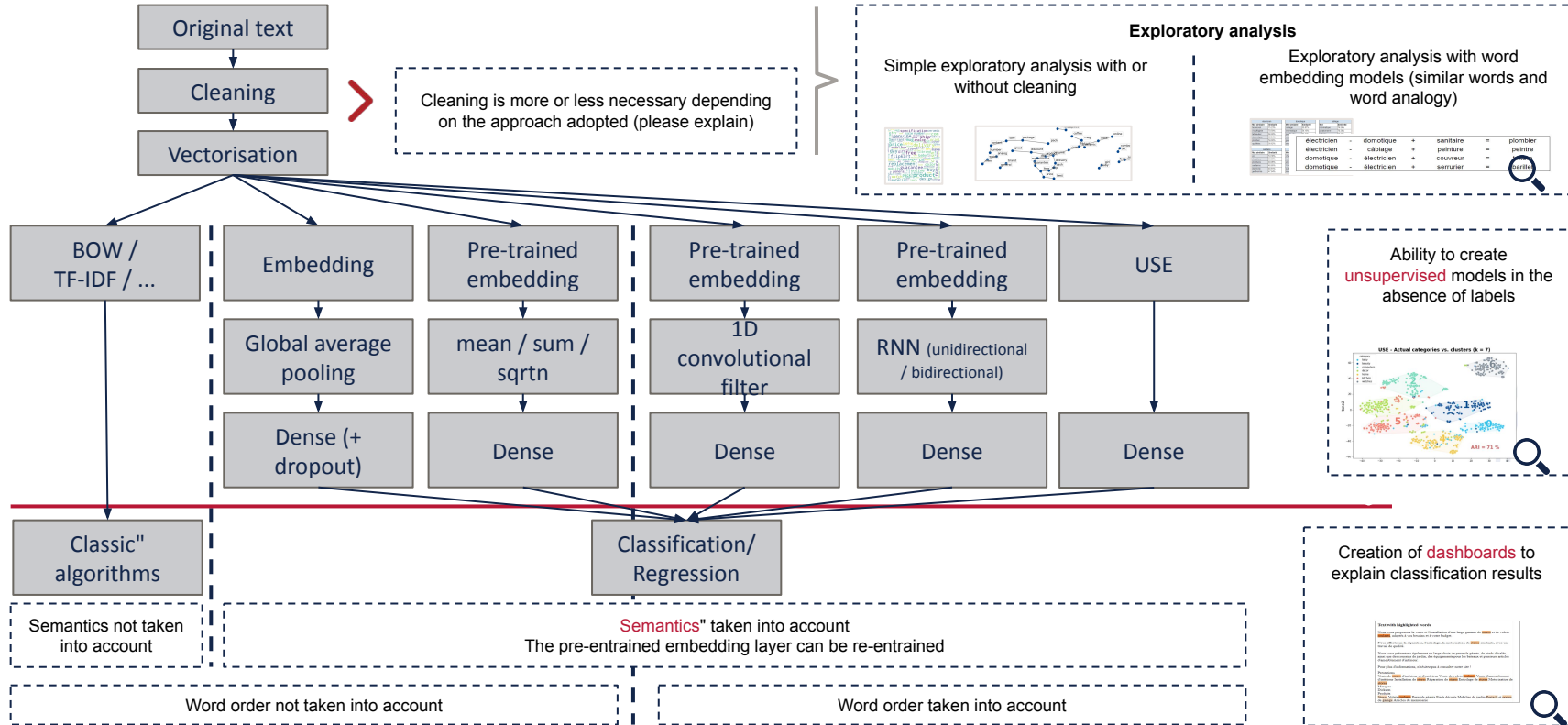
- Presentation of approaches
- Topic modelling and segmentation
- Supervised learning
 - Score prediction
 - Theme prediction

3.1 | Presentation of approaches

Objectives

- Understanding the project and its context
- Understanding the benefits of predictive models

Presentation of modelling approaches



Global approaches

- **Use of unsupervised and deterministic approaches to define topics**
 - LDA to explore possible topics
 - Word2vec to identify similar words
 - At the end, define the themes and associated words. The average score can then be calculated for each theme. Examples of themes: reimbursements (reimbursement, service, payment, etc.), customer service and contacts (email, telephone, customer service, etc.), contract management (take out, contract, termination, etc.).
- **Analysis of feelings**
 - In addition to notes, packages can be used to detect the different sentiments present in comments.
 - To use certain packages for the English language, the comments can be translated into English.
- **Supervised approaches**
 - Word approach: we can
 - Approach using themes: it would also be interesting to predict the score using themes, so that we know which themes have the greatest impact on the score. This would be a classic machine learning approach.

Instructions for defining the themes

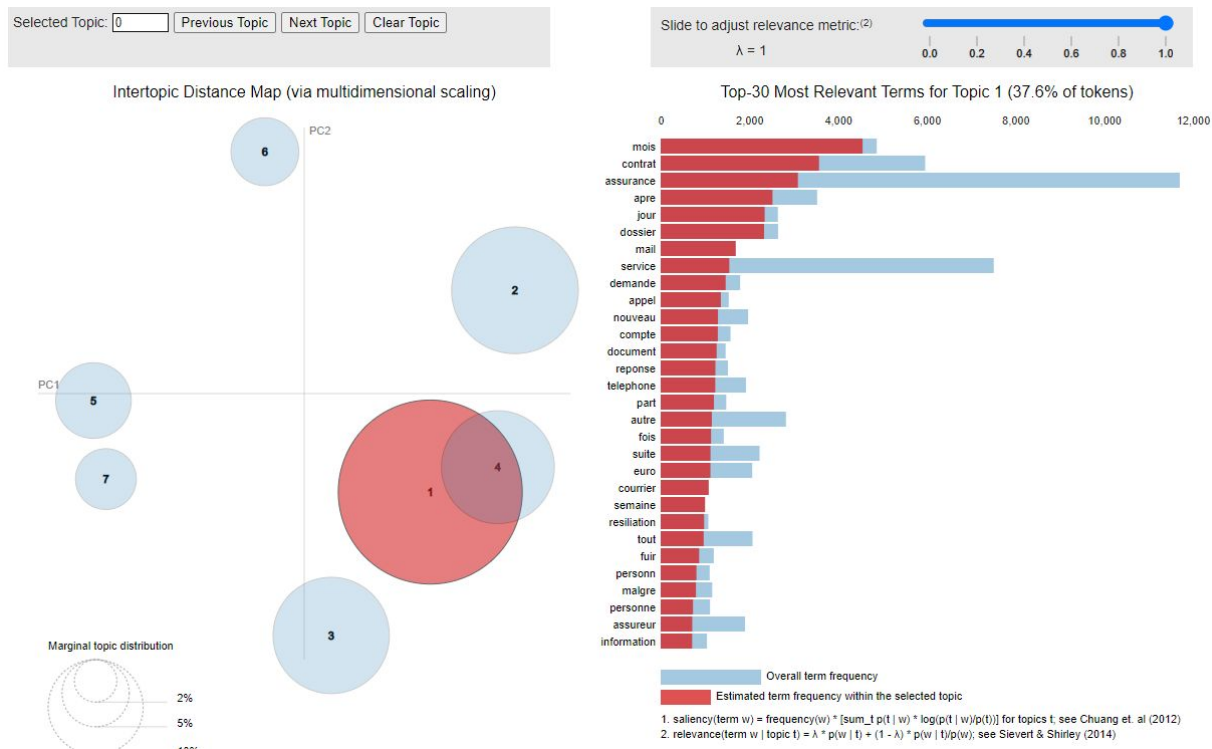
- **List of themes with deterministic keyword detection**
 - We can help with variants using textual distance or word2vec
 - The theme is considered to be associated with a comment as soon as a word is detected.
- **Creation of the learning base**
 - Themes are predicted using supervised models
- **Interpretation of results**

3.2 | Topic modelling and segmentation

Objectives

- Identify topics and associated keywords
- Comment segmentation
- Use embeddings to create a final list of themes with associated keywords

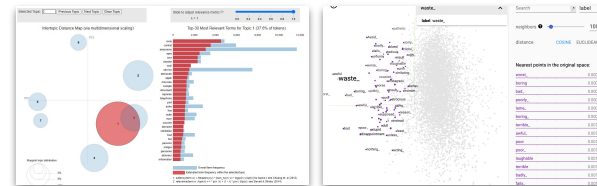
Illustration LDA - Topic modeling



Identifying themes: complements with word2vec

In addition to **topic modelling**, **word2vec** modelling can be used to identify additional keywords.

Initially, these themes can be detected deterministically, then a supervised model will be built to predict them.



- **Price:** price, price, price, expensive...
- **Health reimbursement:** reimbursement, care, dental, optical, glasses...
- **Claims management:** incidents, accidents, disputes, fender-benders, burglaries, etc.
- **Insurance products:** home, car, motorbike...
- **Positive feelings:** perfect, super, top, perfectly...

assurance	
Mot similaire	Similarité
assurances	61%
compagnie	61%
expérience	60%
ami	46%
option	42%

auto	
Mot similaire	Similarité
habitation	88%
automobile	87%
souscris	79%
souscrit	75%
motos	61%

prix	
Mot similaire	Similarité
tarif	92%
tarifs	88%
marché	73%
offre	70%
garanties	70%

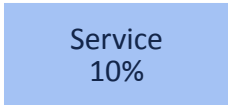
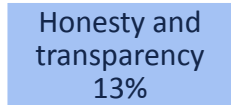
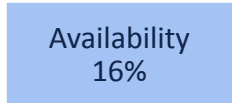
remboursement	
Mot similaire	Similarité
soins	73%
dentaire	69%
honoraires	68%
lunettes	67%
paiement	65%

sinistre	
Mot similaire	Similarité
incident	79%
accident	76%
litige	69%
accrochage	66%
cambrilage	64%

parfait	
Mot similaire	Similarité
super	89%
top	87%
parfaitement	87%
impeccable	87%
sympa	86%

Illustration

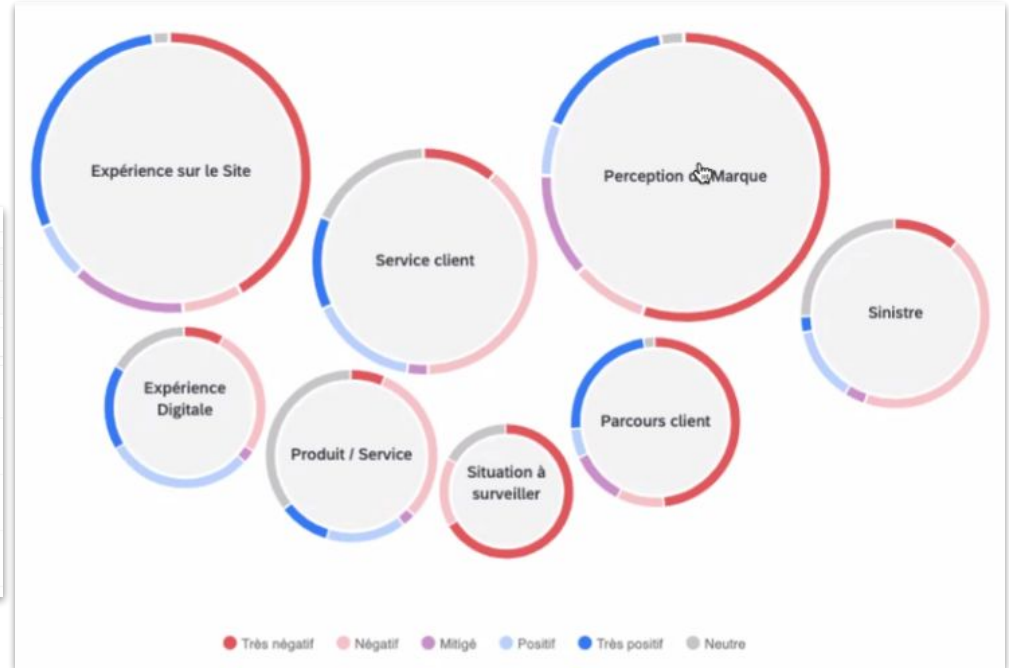
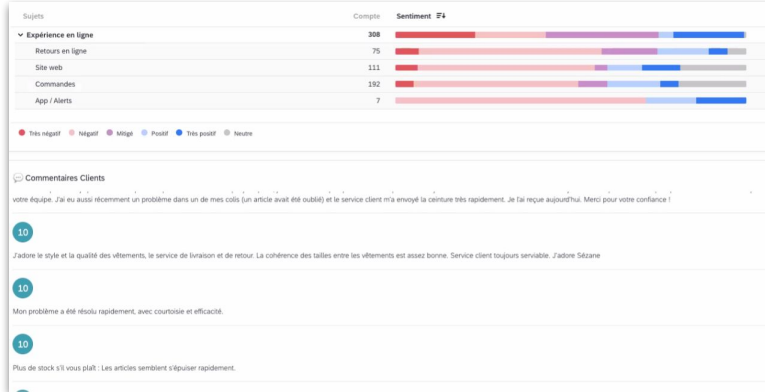
Prices
27%



Analysis of themes according to scores

Illustration

- Create themes that can be fine-tuned as you go along, with keywords to associate with them.
- Prepare answers if necessary



Exploring comments

Each comment can be assigned a theme. Modifications/corrections can be made manually.

Illustration

Tous les commentaires 1459

Catégorisé

0 % Non catégorisés 100 %

Tous les sujets + - ...

Rechercher des sujets

Expérience Digitale 111

Expérience sur le Site ...

Commercialisation ...

Compte

Connexion

Création de compte

Conception

Contenu

Facilité d'utilisation 53

Navigation 151

Parcours client 142

Modifier

Supprimer

Effacer la requête

Ajouter un sujet dessous

Déplacer

Impossible de vous joindre, ni téléphone ni chat, et impossible d'agir sur le site une catastrophe ! contrat auto bien place niveau prix au départ, et a l'occasion d'un changement de véhicule, vous êtes au double

Commercialisation

Conception

Expérience Digitale

Navigation

Prix

Téléphone

Commercialisation

Remarques Générales

Un très bon niveau de conseil ; une qualité de la relation par téléphone excellente. Une rapidité de m

Conception Perception Générale Téléphone

C'est bien dans l'idée de proposer des prix bas mais le reste ne suis pas - des frais de 176 pour un c

Nul

Satisfait du service & de l'assurance souscrit auprès de l'Olivier Assurance. C'est une première auprès

Contenu Perception Générale

La réponse à ma demande a été rapide et correspond à ce que je recherche. Les explications sont cla

Service client

Rien à redire, si ce n'est juste petit accroc à la signature électronique de mon contrat ce jour. La mail c

conservé bien scrupuleusement.

Example of text annotations

Text annotation tools would make it easier to identify the most valuable elements in texts.

Illustration

Ground truth Prediction

Item to be returned
Item to be returned

A tiara... a white gold tiara for a newborn baby. Yeah... you know, I

Customer
Store location

think she got that at Gertrude Zachary's in Nob Hill. I mean that

Price
Price

TOOLS

- 1 Patient
- 2 Symptom
- 3 Condition

Patient

Lucy has a set of medical conditions that are summarised

as HERN (Hereditary Endotheliopathy, Retinopathy,

and Stroke). 10 years ago, she experienced symptoms

Symptom Symptom Symptom Symptom

of proptosis, tachycardia, thirst, and vomiting and was

diagnosed with Grave's Disease. She was treated with

Neomercazole. The coexisting medical condition may be

exerting an impact on Lucy's current anxious state. In

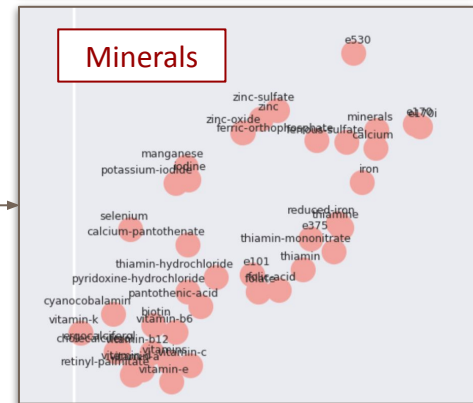
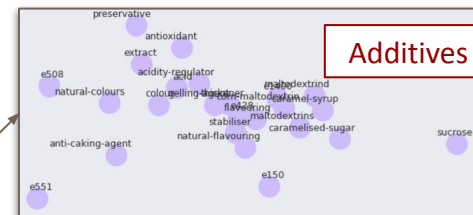
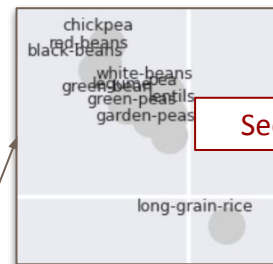
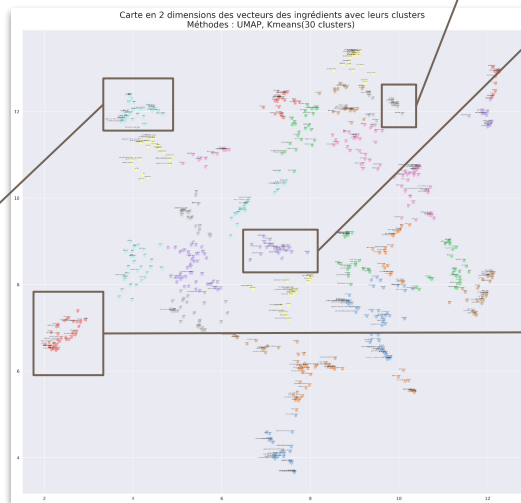
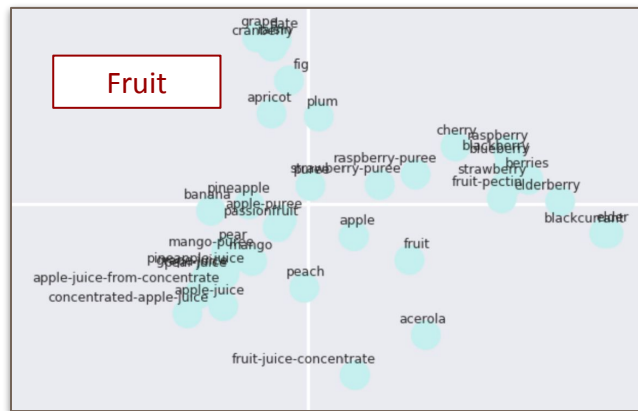
Skip Submit

Example of Food2vec (project 1)

Ingredient embeddings allow similar ingredients to be grouped together using an unsupervised model.

Business expertise is therefore important for checking the consistency of these clusters, and providing labels for each cluster.

(Attempt to label clusters in red)



This study was carried out by Quentin Navarre (M2 ESILV 2022)

Final list and use

- link:
<https://docs.google.com/spreadsheets/d/1B-IF8eyVEq1CWn5D36OCR9OqgUYZWozl/edit?usp=sharing&ouid=108295787782546724708&rtpof=true&sd=true>
- Give an average rating per theme (the average rating of all the reviews that talk about this theme)

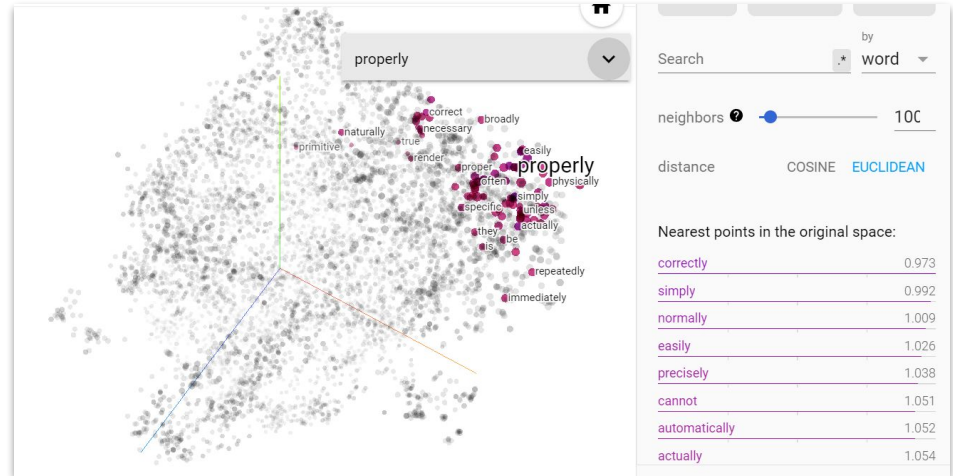
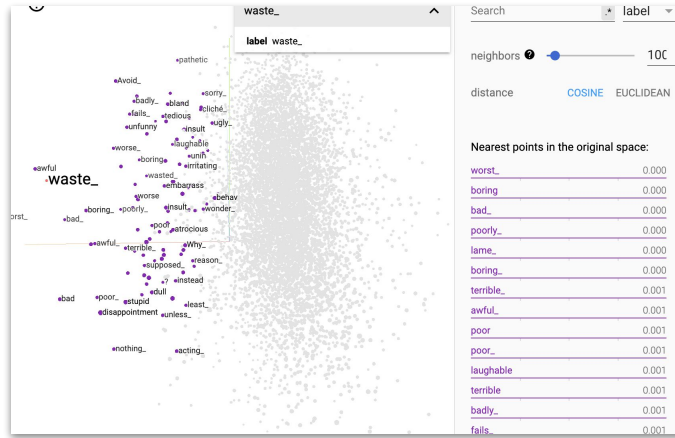
3.3 | Word embedding

Objectives

- Creating word embeddings
- Visualisation
- Applications

Creating embeddings using the self-supervised approach

- Tensorflow offers a plugin that allows you to view embeddings (see the example here <https://projector.tensorflow.org/>).
- You can use this plugin to visualise your own results using the Tensorboard's built-in projector (https://www.tensorflow.org/tensorboard/tensorboard_projector_plugin).



Applications to be carried out

- **Compare several terms**
- **For a term, find similar terms**
- **Word analogy**

Identifying similar professions and analysing skills (1/2)

Illustration

By learning a word embedding model based on bulk texts of construction and public works activities, similar trades and associated similar activities can be highlighted.

Subsequently, embeddings can be created for a nomenclature of activities. Segmentation can be carried out to group together similar occupations.

Semantic similarity of words

electricien	
Mot similaire	Similarité
technicien	75.23%
chauffagiste	70.59%
tableautier	68.80%
domestique	65.04%
plombier	59.89%
qualifelec	58.62%

domotique	
Mot similaire	Similarité
cablage	58.87%
informatique	54.18%
parafoudre	53.29%
videosurveillance	53.06%
generale	49.74%
depannage	49.63%

cablage	
Mot	Similarité
informatique	74.54%
paratonnerre	70.29%
schema	67.32%
parafoudre	67.20%
tableautier	65.52%
disjoncteur	64.41%

sanitaire	
Mot similaire	Similarité
wc	53.36%
chaudiere	50.00%
plomberie	49.96%
sanitaires	48.65%
electricite	48.61%
geothermie	47.64%

plombier	
Mot similaire	Similarité
chauffagiste	74.47%
technicien	62.38%
electricien	59.89%
1964	53.29%
qualifie	53.15%
depanneur	51.52%

metallerie	
Mot	Similarité
ferronnerie	70.57%
serrurerie	58.79%
tolerie	57.98%
chaudronnerie	57.52%
ferronnier	54.72%
metalliques	53.10%

Detection of similar words

Semantic similarity of words

fuir	
Mot	Similarité
fuire	93%
éviter	91%
éviter	90%
fuyez	83%
bannir	81%
honte	79%
déconseiller	76%
absolument	76%
lamentable	75%
honteux	74%

parfait	
Mot	Similarité
nickel	91%
super	89%
top	88%
efficace	88%
clair	88%
accessible	87%
réactif	86%
sympa	85%
impeccable	85%
reactif	84%

electricien	
Mot similaire	Similarité
technicien	75.23%
chauffagiste	70.59%
tableautier	68.80%
domestique	65.04%
plombier	59.89%
qualifelec	58.62%

sanitaire	
Mot similaire	Similarité
wc	53.36%
chaudiere	50.00%
plomberie	49.96%
sanitaires	48.65%
electricite	48.61%
geothermie	47.64%

domotique	
Mot similaire	Similarité
cablage	58.87%
informatique	54.18%
parafoudre	53.29%
videosurveillance	53.06%
generale	49.74%
depannage	49.63%

plombier	
Mot similaire	Similarité
chauffagiste	74.47%
technicien	62.38%
electricien	59.89%
1964	53.29%
qualifie	53.15%
depanneur	51.52%

cablage	
Mot	Similarité
informatique	74.54%
paratonnerre	70.29%
schema	67.32%
parafoudre	67.20%
tableautier	65.52%
disjoncteur	64.41%

metallerie	
Mot	Similarité
ferronnerie	70.57%
serrurerie	58.79%
tolerie	57.98%
chaudronnerie	57.52%
ferronnier	54.72%
metalliques	53.10%

The word2vec model is used to find similar words in relation to reference words. This makes it possible to create a segmentation (or themes).

4 | Supervised learning

4.1 | Classification - theme prediction

Objectives

- Understanding the project and its context
- Understanding the benefits of predictive models

Methodological approach

- **In the previous section, a list of themes was defined with associated keywords. This was a text annotation phase. For each comment, we can then detect the presence of this theme by checking for the presence of keywords. But there may be false positives or false negatives.**
 - False positives: a keyword is not enough to detect the theme
 - False negatives: the associated keywords have not been identified to detect this theme
- **To remedy this problem, you can train a model with the labelled base, then reapply this model.**
 - We can examine the high scores for which the label was set to negative. After checking, we can confirm whether these are false negatives and change the label.
 - The same is done for positive classes with low scores.
 - Why might this process work?

Classical machine learning - classification and regression

Text pre-processing

- https://scikit-learn.org/stable/modules/generated/sklearn.feature_extraction.text.TfidfVectorizer.html

Machine learning models

- <https://pypi.org/project/lazypredict/>

Error analysis

- Display and analyse the observations for which the model was most wrong.

4.2 | Score prediction

Objectives

- Understanding the project and its context
- Understanding the benefits of predictive models

Text classification with Tensorflow

Examples

- <https://www.kaggle.com/code/jeremiahharmesen/tensorflow-hub-for-text-classification>
- <https://github.com/mohamedELBAHA/Text-classification-of-BBC-news-dataset>

Error analysis

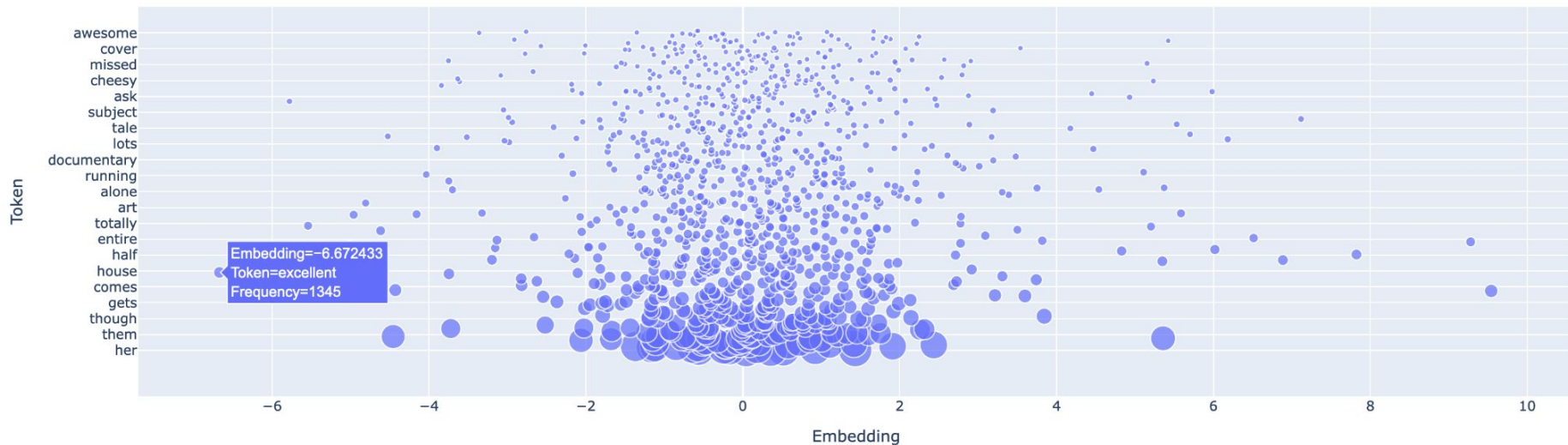
- Display and analyse the observations for which the model was most wrong.

Sentiment classification of IMBD opinions (1/)

Illustration

Both polarities can be observed with embeddings

With only one dimension embedding

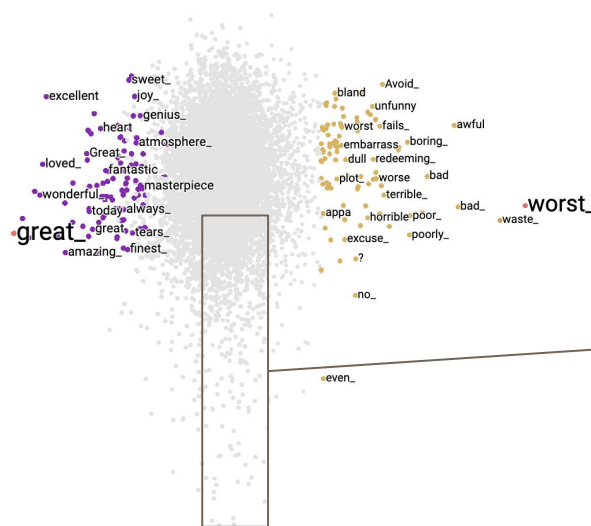


Sentiment classification of IMBD opinions (1/)

Illustration

Both polarities can be observed with embeddings

distance	COSINE	EUCLIDEAN
Nearest points in the original space:		
perfect	0.203	
best_	0.313	
excellent_	0.419	
wonderful_	0.501	
loved_	0.568	
excellent	0.629	
favorite_	0.813	
perfect_	0.953	
amazing_	0.961	
enjoyed_	1.090	
wonderful	1.105	
fantastic_	1.282	



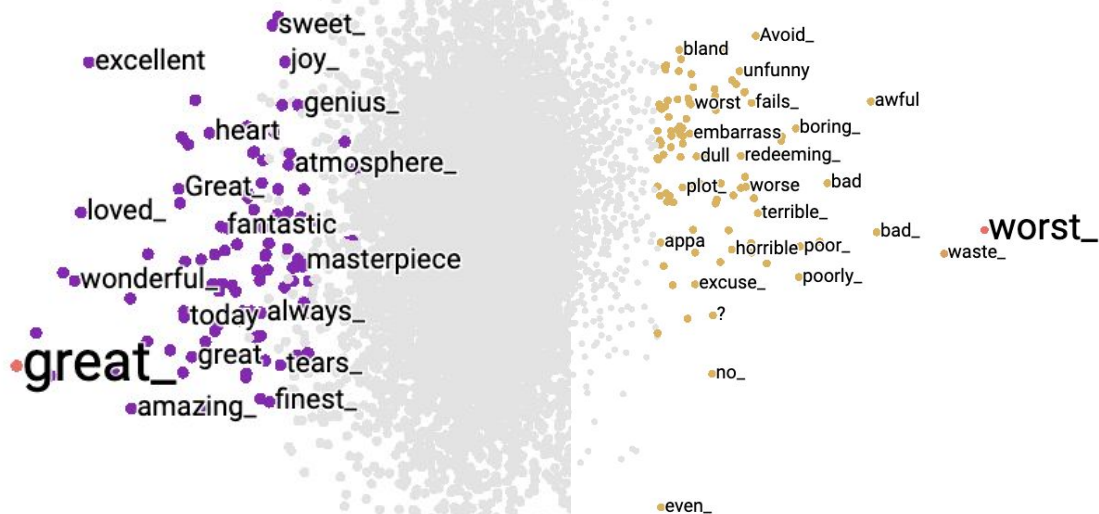
distance	COSINE	EUCLIDEAN
Nearest points in the original space:		
waste_	0.526	
bad_	1.362	
awful	1.435	
bad	1.977	
awful_	2.070	
poor_	2.317	
poorly_	2.330	
boring_	2.371	
terrible	2.547	
worse_	2.552	
nothing_	2.735	
terrible_	2.845	

unknown #91

distance	COSINE	EUCLIDEAN
Nearest points in the original space:		
for	0.172	
ution_	0.174	
character	0.195	
in_	0.203	
Mad	0.213	
mod	0.213	
ig	0.218	
fast_	0.247	
defeat	0.252	
which_	0.257	
passed_	0.262	
<	0.262	
Indian	0.263	
determine	0.271	
work	0.276	

Sentiment

Both polarities



Illustration

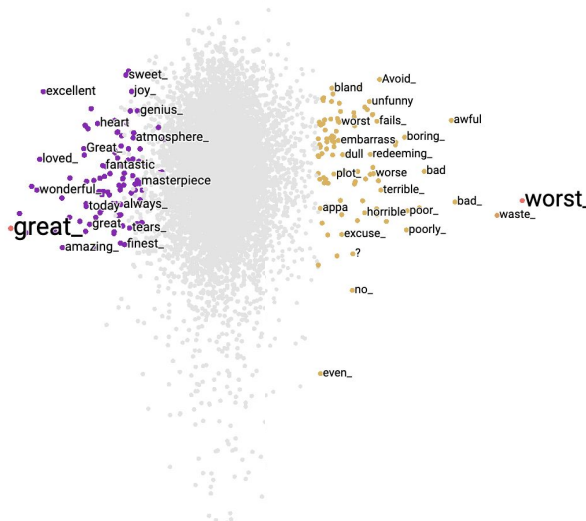
Sentiment classification of IMBD opinions (1/)

Illustration

Both polarities can be observed with embeddings

distance	COSINE	EUCLIDEAN
Nearest points in the original space:		
perfect	0.001	
enjoyed_	0.001	
today	0.001	
best_	0.001	
wonderful_	0.001	
perfectly_	0.001	
great	0.001	
excellent_	0.001	
definitely_	0.001	
perfect_	0.001	
favorite_	0.001	
masterpiece	0.001	

distance	COSINE	EUCLIDEAN
Nearest points in the original space:		
perfect	0.203	
best_	0.313	
excellent_	0.419	
wonderful_	0.501	
loved_	0.568	
excellent	0.629	
favorite_	0.813	
perfect_	0.953	
amazing_	0.961	
enjoyed_	1.090	
wonderful	1.105	
fantastic_	1.282	



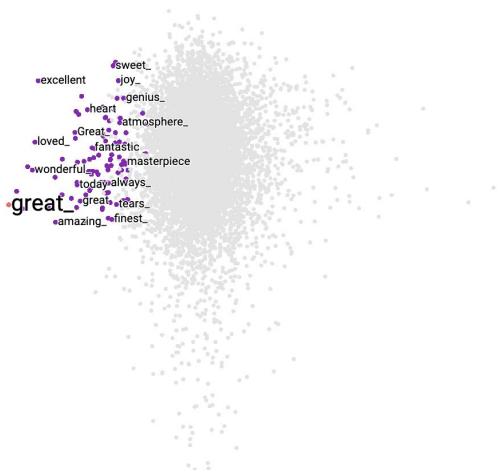
distance	COSINE	EUCLIDEAN
Nearest points in the original space:		
waste_	0.000	
poorly_	0.000	
awful_	0.000	
boring_	0.000	
ridiculous_	0.000	
awful	0.000	
fails_	0.001	
terrible_	0.001	
bad	0.001	
bad_	0.001	
Avoid_	0.001	
poor_	0.001	

distance	COSINE	EUCLIDEAN
Nearest points in the original space:		
waste_	0.526	
bad_	1.362	
awful	1.435	
bad	1.977	
awful_	2.070	
poor_	2.317	
poorly_	2.330	
boring_	2.371	
terrible	2.547	
worse_	2.552	
nothing_	2.735	
terrible_	2.845	

Example: classification of the IMBD sentiment database

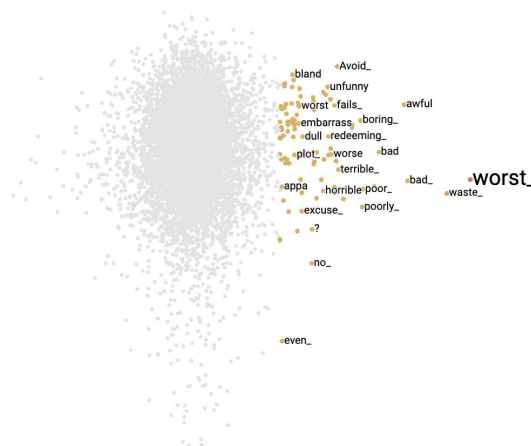
Illustration

Both polarities can be observed
with embeddings



distance	COSINE	EUCLIDEAN	distance	COSINE	EUCLIDEAN
Nearest points in the original space:			Nearest points in the original space:		
perfect	0.001		perfect	0.203	
enjoyed_	0.001		best_	0.313	
today	0.001		excellent_	0.419	
best_	0.001		wonderful_	0.501	
wonderful_	0.001		loved_	0.568	
perfectly_	0.001		excellent	0.629	
great	0.001		favorite_	0.813	
excellent_	0.001		perfect_	0.953	
definitely_	0.001		amazing_	0.961	
perfect_	0.001		enjoyed_	1.090	
favorite_	0.001		wonderful	1.105	
masterpiece	0.001		fantastic_	1.282	

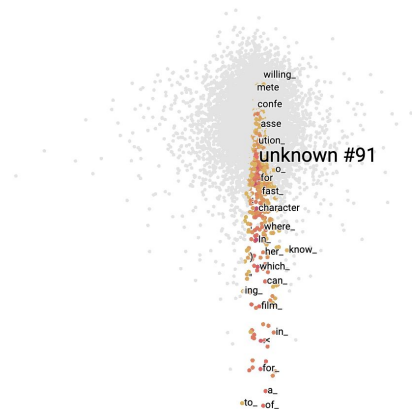
Analysis of embeddings that contribute to negative sentiment.



distance	COSINE	EUCLIDEAN
Nearest points in the original space:		
waste_	0.000	
poorly_	0.000	
awful_	0.000	
boring_	0.000	
ridiculous_	0.000	
awful	0.000	
fails_	0.001	
terrible_	0.001	
bad	0.001	
bad_	0.001	
Avoid_	0.001	
poor_	0.001	

distance	COSINE	EUCLIDEAN
Nearest points in the original space:		
waste_	0.526	
bad_	1.362	
awful	1.435	
bad	1.977	
awful_	2.070	
poor_	2.317	
poorly_	2.330	
boring_	2.371	
terrible	2.547	
worse_	2.552	
nothing_	2.735	
terrible_	2.845	

Terms that don't have much impact



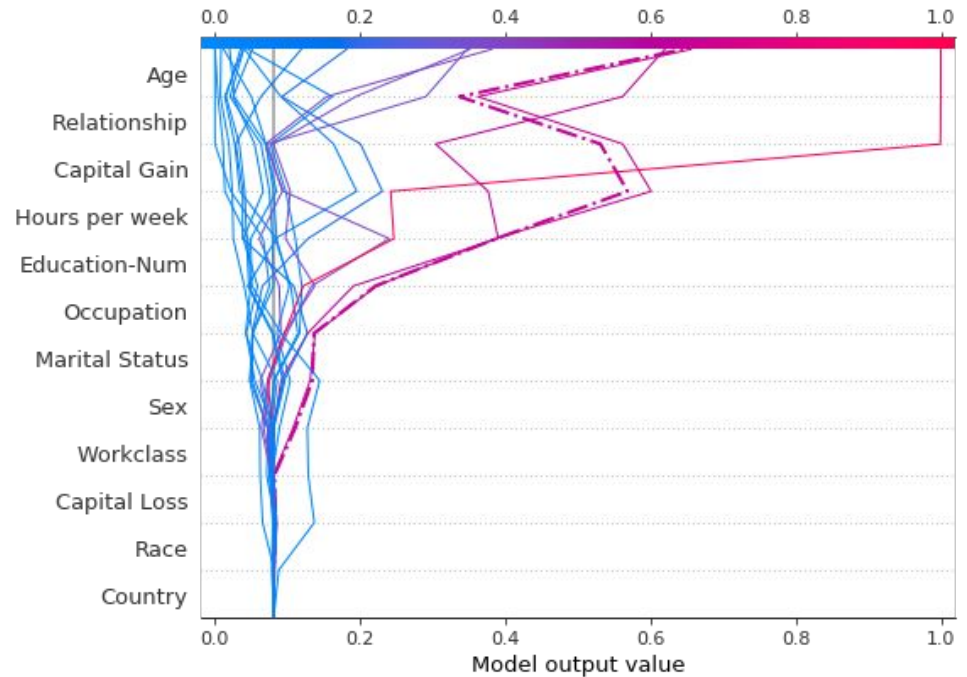
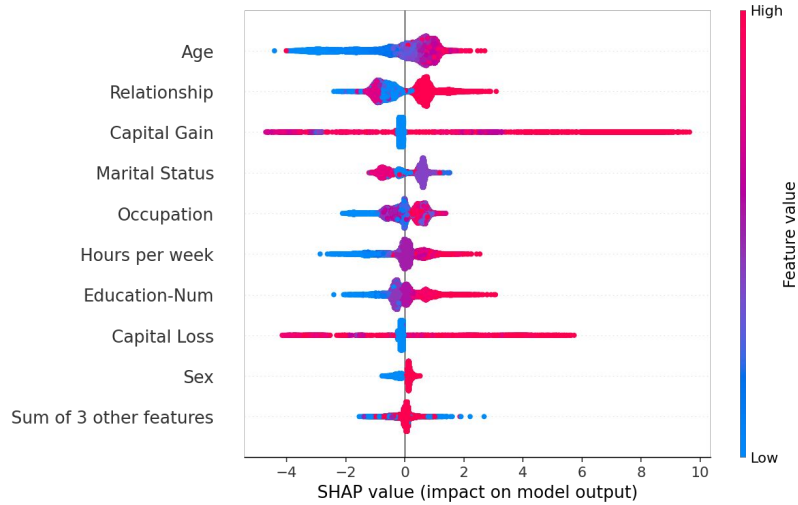
distance	COSINE	EUCLIDEAN
Nearest points in the original space:		
for	0.172	
ution_	0.174	
character	0.195	
In_	0.203	
Mad	0.213	
mod	0.213	
ig	0.218	
fast_	0.247	
defeat	0.252	
which_	0.257	
passed_	0.262	
.<	0.262	
Indian	0.263	
determine	0.271	
work	0.276	

4.3 | Model interpretation

Objectives

- Themes explaining the level of satisfaction
- Keywords/embeddings that explain the level of satisfaction

Shap: tabular data



Shap: text plot

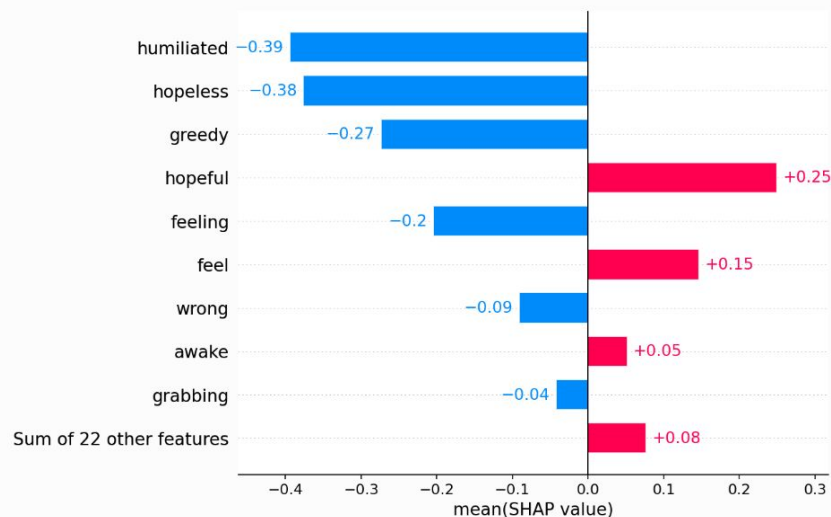
https://shap.readthedocs.io/en/latest/example_notebooks/api_examples/plots/text.html



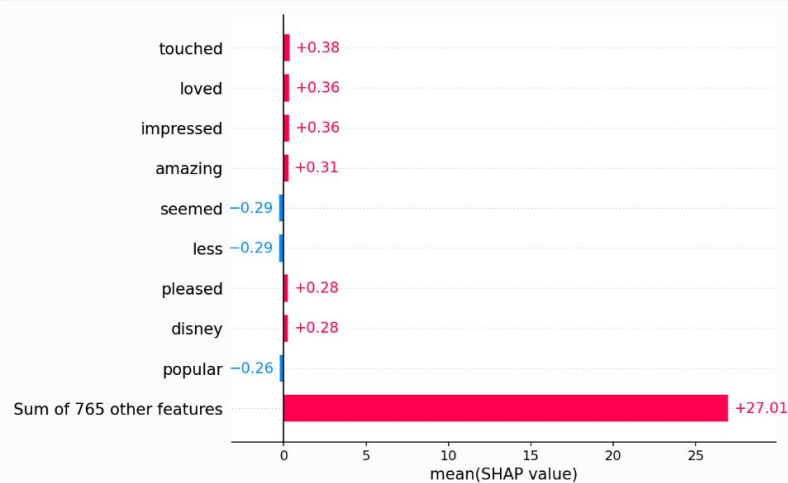
Shap: sentiment analysis

https://shap.readthedocs.io/en/latest/text_examples.html

```
shap.plots.bar(shap_values[:, :, "joy"].mean(0))
```



```
shap.plots.bar(shap_values[:, :, "POSITIVE"].mean(0))
```



5

Streamlit applications

- Themes explaining the level of satisfaction
- Keywords/embeddings that explain the level of satisfaction

5 | Appendices

- Templates



Title

- Text
 - Text
- Text
 - Text



Title

- Text
 - Text
- Text
 - Text



Title

- Text
 - Text
- Text
 - Text

Kicker

Title

- **Level 1**
 - Level 2
 - Level 2
 - Level 3
- **Level 1**
 - Level 2
 - Level 2
 - Level 3

Title

- **Level 1**
 - Level 2
 - Level 2
 - Level 3
- **Level 1**
 - Level 2
 - Level 2
 - Level 3

Title

- **Level 1**
 - Level 2
 - Level 2
 - Level 3
- **Level 1**
 - Level 2
 - Level 2
 - Level 3

Kicker

Title

- **Level 1**
 - Level 2
 - Level 2
 - Level 3
- **Level 1**
 - Level 2
 - Level 2
 - Level 3

Title

- **Level 1**
 - Level 2
 - Level 2
 - Level 3
- **Level 1**
 - Level 2
 - Level 2
 - Level 3

Title

- **Level 1**
 - Level 2
 - Level 2
 - Level 3
- **Level 1**
 - Level 2
 - Level 2
 - Level 3

Title

- **Level 1**
 - Level 2
 - Level 2
 - Level 3
- **Level 1**
 - Level 2
 - Level 2
 - Level 3

Title

Text

Text

Text

Detail

- **Text 1**
 - Text
- **Text 1**
 - Text
- **Text 1**
 - Text
- **Text 1**
 - Text
- **Text 1**
 - Text
- **Text 1**
 - Text

Kicker

| Chart title

Subtitle

Source

Text



	Title	Title	Title	Title	Title
	Text	Text	Text	Text	Text
Text	▪Text	▪Text	▪Text	▪Text	▪Text
Text	▪Text	▪Text	▪Text	▪Text	▪Text
Text	▪Text	▪Text	▪Text	▪Text	▪Text
Text	▪Text	▪Text	▪Text	▪Text	▪Text
Text	▪Text	▪Text	▪Text	▪Text	▪Text
Text	▪Text	▪Text	▪Text	▪Text	▪Text

Text	Text	Text	Text	Text	Text	Text
Text	▪ Text	▪ Text	▪ Text	▪ Text	▪ Text	▪ Text
Text	▪ Text	▪ Text	▪ Text	▪ Text	▪ Text	▪ Text
Text	▪ Text	▪ Text	▪ Text	▪ Text	▪ Text	▪ Text
Text	▪ Text	▪ Text	▪ Text	▪ Text	▪ Text	▪ Text
Text	▪ Text	▪ Text	▪ Text	▪ Text	▪ Text	▪ Text
Text	▪ Text	▪ Text	▪ Text	▪ Text	▪ Text	▪ Text

Procedure sheet



OBJECTIVES

- Text



DELIVERABLES

- Text



CONTRIBUTORS

- Text



WORK TO BE CARRIED OUT

- **Text**
 - Text
 - Text

- **Text**
 - Text
 - Text

- **Text**
 - Text
 - Text

- **Text**
 - Text
 - Text

- **Text**
 - Text
 - Text

- **Text**
 - Text
 - Text

< Mission title > (1 page reference sheet)

Customer
logo

OR

CONFIDENTIAL

CONTEXT AND OBJECTIVES

- **Explain the context in which Exton intervened**
 - Text
- **What were the objectives and challenges?**
 - Text
- **Who were the decision-makers/stakeholders?**
 - Text

APPROACH

- **Present the project approach, possibly with a graph**
 - The sites, the stages, the schedule
 - Animation methods
 - Any governance/decision-making bodies to which the project has been presented

RESULTS

- **Present the results of the project**
 - Text
 - Text
- **Illustrate with examples of typical deliverables (vignettes)**
 - Text
 - Text